# Modelling Pollinator Niches, Plant–Pollinator Coupling, and Community Turnover

Species Distribution Modelling (SDM) and Generalized Dissimilarity Modelling (GDM)

**FELAB | Konkuk University (Data: National Ecosystem Survey 3rd–5th)**

November 2026 | Seoul, South Korea

# Introduction: From Species Niches to Ecological Interactions

- **Scope**: 2050 climate and land-use impacts across three ecological scales:
  - **Species**: Habitat suitability, realized thermal limits, and 2050 range shifts
  - **Interactions**: Spatial co-occurrence and mismatch with floral and host plants
  - **Communities**: High-elevation species replacement vs. lowland nested attrition
- **Workflow**: Presence records $\rightarrow$ 2050 projections $\rightarrow$ Plant–pollinator overlap $\rightarrow$ Stacked SDM (SSDM) + GDM

| Step | Focus | Core Ecological Topic | Notebook Section & Slides |
|---|---|---|---|
| **Step 1** | **Niches & Drivers** | 5 orders, 12 groups (**FG01–FG12**), 22 predictors | `Step 1–A / 1–B` (Slides 4–13) |
| **Step 2** | **SDM & Climate** | MaxEnt SDM, spatial block CV, & `Bio5` masks | `Step 2–A / 2–B` (Slides 14–25) |
| **Step 3** | **Habitat Coupling** | Plant–pollinator refugia & mismatch deficits | `Step 3` (Slides 26–30) |
| **Step 4** | **Beta Turnover** | Stacked SDM vs. GDM & $\beta_{sim}$ / $\beta_{sne}$ partitioning | `Step 4–A / 4–B` (Slides 31–40) |

# Hands-On Lab: Structure & Access Guide

- **Interactive Notebook (`SDM_HandsOn_Workshop_KO.ipynb`)**:
  - **[실습] Single-Case**: Step-by-step exercise on **`FG03` *Bombus*** (`bioclim1` map, 50 km block CV, `Bio5` mask, *Bombus* × Spring Woody overlap, & 2-site Baselga $\beta$)
  - **[전체 비교] Synthesis**: `scripts/sdm_core.py` across **5 orders**, **12 groups (`FG01`–`FG12`)**, **8 plant–pollinator pairs**, and **4 community matrices**
- **How to Connect**:
  - **`MyBinder`**: **github.com/alanbseo/SDMworkshop** → **`Launch Binder`** badge *(direct: **mybinder.org/v2/gh/alanbseo/SDMworkshop/main**)*
  - **Local PC**: `git clone https://github.com/alanbseo/SDMworkshop.git` → `pip install -r requirements.txt` → `jupyter lab SDM_HandsOn_Workshop_KO.ipynb`

# STEP 1. Ecological Niches & Environmental Drivers

- **Scope & Data Structure**
  - 5 insect orders and 12 functional groups (**FG01–FG12**) across 84,576 inland 1 km cells
  - Presence-only (PO) records and 10,000-cell presence–background (PB) calibration regarding imperfect detection
  - 22 climatic, water-balance, topo-soil, and 100 m land-cover predictors (2020 & 2050)
- **Objective**: Quantify how macroclimate and 100 m landscape structure differentiate pollinator niches.

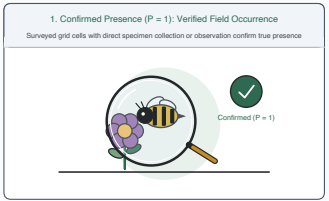
# [Step 1] Pollinator Species and 12 Functional Groups (FG01–FG12)

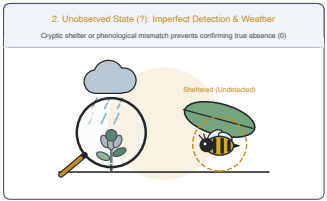
- **Grouping Criteria**: 12 groups defined by within-order sample size and ecological heterogeneity (`Bio5` = realized upper cutoff).

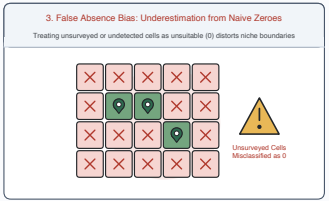
| Order | Functional Group | Ecological Niche & Contrast | Cells | `Bio5` Limit |
|---|---|---|---|---|
| **Hymenoptera** (벌목) | **FG03** *Bombus* (뒤영벌속) | Cold-adapted montane bumblebees | 927 | 33.4 °C |
| **Hymenoptera** (벌목) | **FG04** *Apis* (꿀벌속) | Heat-tolerant eusocial honeybees | 2,352 | 41.5 °C |
| **Hymenoptera** (벌목) | **FG05** Other Apidae (기타 야생꿀벌과) | Carpenter & cuckoo wild bees | 1,343 | 35.1 °C |
| **Hymenoptera** (벌목) | **FG06** Solitary Bees (단독성 야생벌류) | Spring-active solitary wild bees | 1,158 | 33.4 °C |
| **Diptera** (파리목) | **FG01** Syrphidae (꽃등에과) | Widespread generalist hoverflies | 2,753 | 34.7 °C |
| **Diptera** (파리목) | **FG02** Bombyliidae (재니등에과) | Open-habitat parasitoid bee flies | 519 | 33.4 °C |
| **Hemiptera** (노린재목) | **FG07** Temp. Hemiptera (온대 노린재류) | Mainland flower & plant bugs | 2,025 | 33.4 °C |
| **Hemiptera** (노린재목) | **FG08** S. Largidae (큰별노린재과) | Subtropical coastal *Physopelta* | 124 | 35.9 °C |
| **Lepidoptera** (나비목) | **FG09** Papilionidae (호랑나비과) | Forest-interior swallowtails | 535 | 33.3 °C |
| **Lepidoptera** (나비목) | **FG10** Nymphalidae (네발나비과) | Forest-edge brush-footed butterflies | 421 | 35.1 °C |
| **Lepidoptera** (나비목) | **FG11** Other Lepidoptera (기타 나비·나방류) | Open/meadow butterflies & moths | 296 | 37.7 °C |
| **Coleoptera** (딱정벌레목) | **FG12** Cerambycidae (하늘소과) | Forest xylophagous flower longhorns | 782 | 33.2 °C |

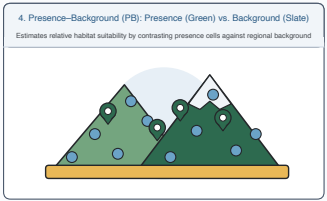
# [Step 1] Imperfect Detection in National Survey Records

- **Unobserved** $\neq$ **Absent**: Survey gaps and weather create false absences (?), motivating presence–background (PB) modelling (Elith et al., 2011; Guillera-Arroita et al., 2015).



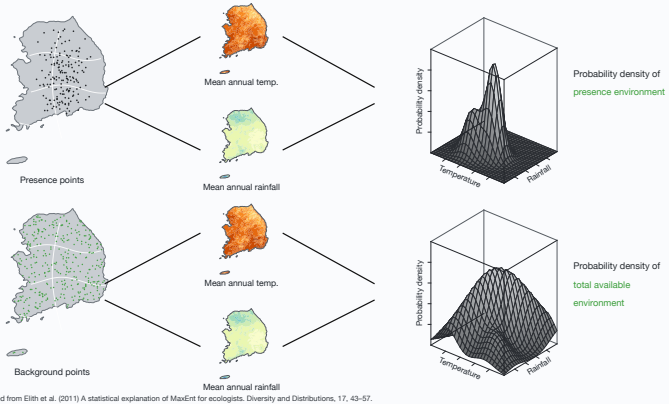



# [Step 1] Presence–Background (PB) Niche Calibration (PO → MaxEnt)

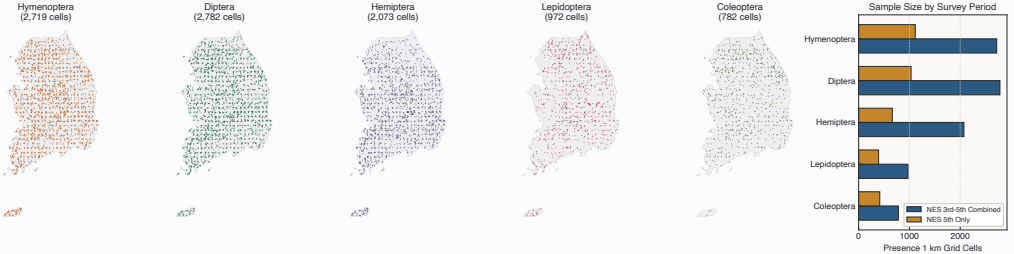
- **Density Ratio ($f_1(\mathbf{z})/f(\mathbf{z})$)**: Contrasts predictor densities at **presence sites** against the **1 km background** (Phillips & Dudík, 2008; Elith et al., 2011).




Adapted from Elith et al. (2011) A statistical explanation of MaxEnt for ecologists. Diversity and Distributions, 17, 43–57.

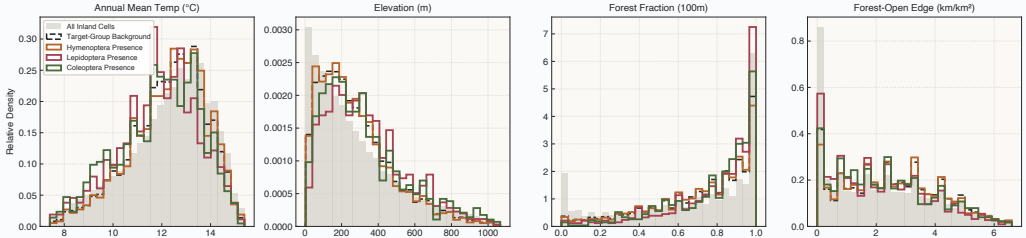
# [Step 1] Biogeography and Survey Coverage Across Five Orders

- **Survey Coverage**: **4,523** surveyed 1 km cells (5.3% of 84,576 inland cells), led by Diptera (2,782) and Hymenoptera (2,719).



# [Step 1] Forest-Edge Selection vs. Survey Sampling Bias

- **Survey Bias vs. Niche Selection**: Surveyed cells (`edge_F0`: 2.54 vs. 2.19 km/km² inland; Phillips et al., 2009) favor forest edges; **Hymenoptera** (2.56) and **Hemiptera** (2.62) select edge mosaics, whereas **Lepidoptera** (2.15) and **Coleoptera** (2.33) shift to closed forest (`frac_2`: 0.78–0.79).



# [Step 1] Baseline Thermal and Habitat Niche Differentiation

- **Composite Habitat Profiles**: Standardized occupied-cell means separate cold-montane **FG03** ***Bombus*** from generalist **FG04** ***Apis*** and coastal **FG08**.

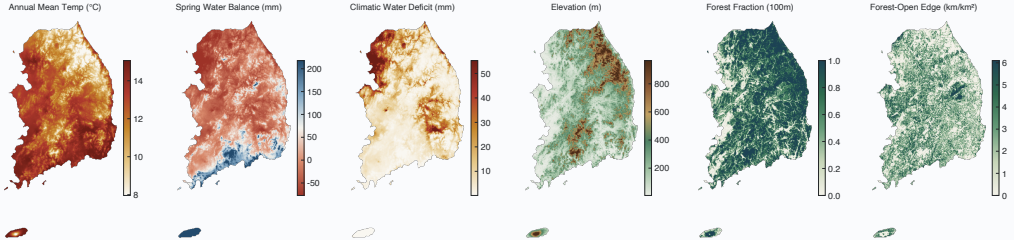
# [Step 1] Input Data and Predictors: SDM vs. GDM

- **Shared vs. Specific Inputs**: Share **22 core 1 km rasters**, but differ in **biological response**, **masks**, and **spatial distance**.

| Category | Variables / Codes | Model | Input Representation & Role |
|---|---|---|---|
| **Response** | Presence-only (PO) vs. Site × Spp. | **SDM** $\neq$ **GDM** | Presence–background (PB, $y_i$) vs. pair $d_{ij}$ |
| **Bioclimate (5)** | `bioclim1, 2, 3, 13, 14` | **SDM & GDM** | Cell $x_i$ (L/Q/H) vs. pair $\lvert\Delta I_k(x)\rvert$ |
| **Atmosphere (3)** | `solarradiation, RH, VPD` | **SDM & GDM** | Solar radiation, humidity, vapor pressure deficit |
| **Water Bal. (3)** | `AI, WB_spr, CWD` | **SDM & GDM** | Aridity index, spring moisture, water deficit |
| **Topo & Soil (5)** | `elev, slope, soildeep3..stone` | **SDM & GDM** | Relief & soil depth/drainage/gravel |
| **Land Cover (6)** | `frac_0..3, edge_FO, shannon` | **SDM & GDM** | 100 m cover, forest–open edge, diversity |
| **Constraints** | `bioclim5, 6, frac_6` | **SDM** | Occupied 99th/1st %ile ($P_{99}/P_{01}$) & water |
| **Spatial** | `Geo_Distance` (`x, y` km) | **GDM** (SDM CV) | GDM spline $I_{\text{geo}}(d_{ij})$; SDM 50 km CV |

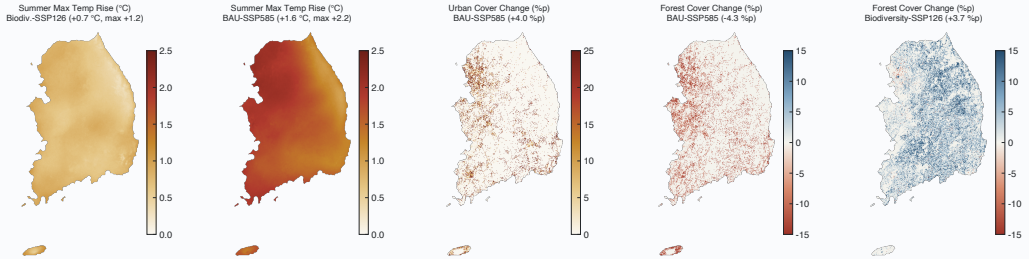
# [Step 1] Baseline Environmental Gradients: Montane Refugia vs. Plains

- **2020 Baseline**: Cool, forested Baekdudaegan mountains vs. warm, drier lowland urban and cropland plains.



# [Step 1] 2050 Climate and Land-Use Scenarios

- **Three Scenarios**: **BAU-SSP585** (+1.7 °C; `Bio5` +1.6 °C mean, max +2.2 °C; urban +4.0%p, forest -4.3%p) · **Climate-SSP126** (+0.6 °C; afforestation +3.4%p) · **Biodiversity-SSP126** (+0.6 °C; sprawl +0.3%p, forest +3.7%p).



# STEP 2. Species Distribution & Climate Change

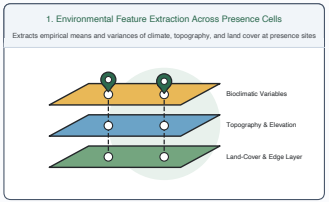
- **Calibration, Validation, & 2050 Projection**
  - Prevalence-balanced L1/L2 PB SDM on MaxEnt L/Q/H features ($\beta_{\text{reg}} = 1.5$; Phillips & Dudík, 2008; Renner & Warton, 2013)
  - **50 km × 50 km spatial block CV** (Roberts et al., 2017; AUC, TSS [Allouche et al., 2006]) & realized `Bio5` masks (Araújo et al., 2013)
  - 2050 habitat loss, gain, and elevational/latitudinal shifts across **FG01–FG12**

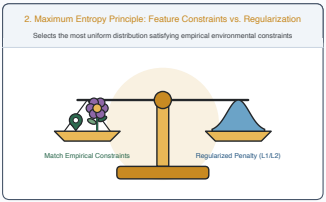
```
model = train_and_validate_sdm(
    pres_ids, env_2020, bg_grid, beta_reg=1.5, use_hinge=True, n_folds=3
)
```

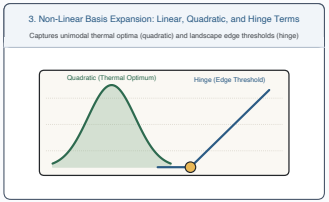
- **Objective**: Identify climate-vulnerable functional groups and persistent montane/coastal refugia.

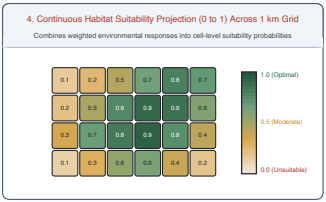
# [Step 2] Capturing Thermal Optima and Habitat Thresholds

- **MaxEnt Basis Expansion**: Fits standardized Linear, Quadratic (thermal optima), and Hinge (edge thresholds) features under L1/L2 regularization (Phillips & Dudík, 2008).



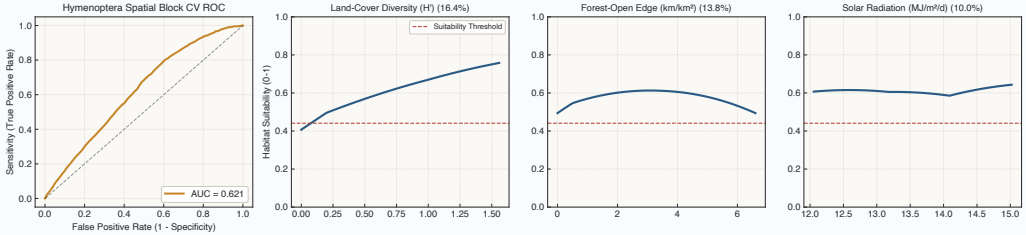



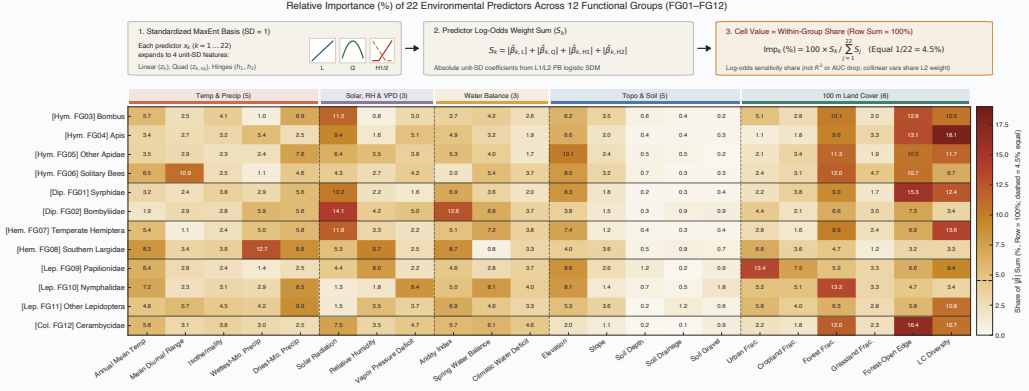
# [Step 2] Response Curves: Edge & Mosaic Dependence in Bees

- **Hymenoptera Diagnostics**: Spatial CV **AUC = 0.621** (TSS = 0.195, $\tau$ = 0.441); standardized $|\hat{\beta}|$ share led by **Diversity (16.4%)**, **Edge (13.8%)**, and **Solar (10.0%)**.



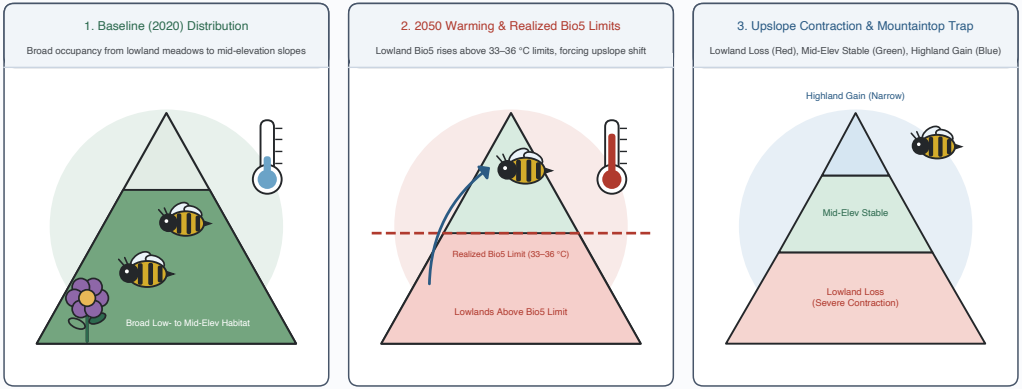
# [Step 2] Variable Importance: Edge Mosaic vs. Forest Specialists

- **Metric**: Row share of unit-SD L/Q/H $|\hat{\beta}|$ sums (row $=$ 100%, equal = **4.5%**).
- **Group Controls (22 Predictors)**: Forest–open edge and diversity (edge_F0, shannon) lead **FG01 Syrphidae** (27.7%), **FG04 *Apis*** (31.2%), and **FG12** (27.1%); **FG10** tracks Forest Fraction (13.2%).



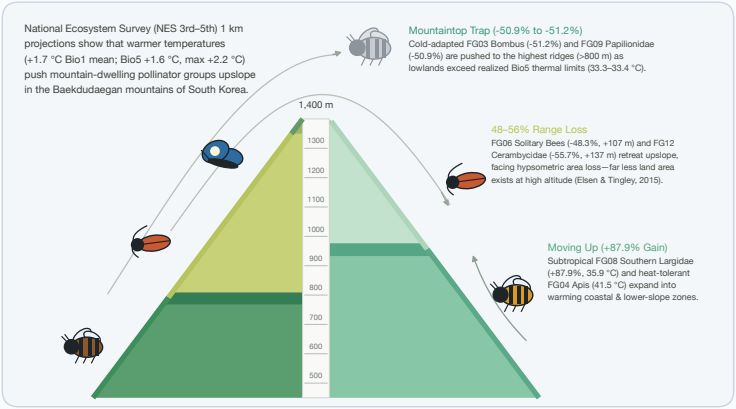
# [Step 2] Lowland Heat Stress and Realized Thermal Limits

- **Summer Warming**: Where 2050 summer maxima (`Bio5`) exceed realized upper niche limits (33–36 °C; Araújo et al., 2013), lowland habitats contract upslope.



# [Step 2] Lowland Loss, Upslope Contraction, and Mountaintop Traps

- **Hypsometric Mountaintop Trap**: Shrinking high-elevation area (Colwell et al., 2008; Elsen & Tingley, 2015) compresses cold-montane **FG03, FG09, FG12**, while **FG08** expands.



# [Step 2] Realized Thermal Cutoffs: Global GBIF and National NES

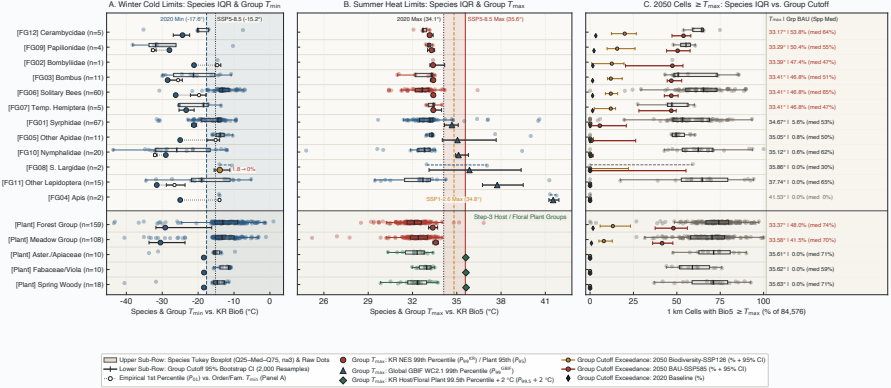
- **Rationale**: National 99th %iles cap warm niches at Korea's ceiling ($P_{99}^{\text{KR}} \leq 33.5$ °C), whereas **WorldClim 2.1** (`10'`, 1970–2000; Fick & Hijmans, 2017) is **3.30 °C cooler** than 1 km **KMA** (2021–2025) and truncates $> 5\%$ of Korean presences.

| Cutoff Source | Occurrence & Climate Grid | Occupied-Cell Percentile Rule |
|---|---|---|
| **1. Global ($P_{99}^{\text{GBIF}}$, $P_{01}^{\text{GBIF}}$)** | **GBIF** (233 poll./274 pl.) + **WC 2.1** (`10'`) | `10'`-thinned 99th/1st %ile of `Bio5/Bio6` |
| **2. National ($P_{99}^{\text{KR}}$, $P_{01}^{\text{KR}}$)** | **NES 3rd–5th** + **KMA** (1 km, 2021–25) | 1 km PO 99th/1st %ile ($\Delta$Bio5 = +3.30 °C) |
| **3. Merged Cutoff** | `Thermal_Limits.json` (Bio5/6) | $T_{\text{max}} = \max(P_{99}^{\text{KR}}, P_{99}^{\text{GBIF}})$; buffered $T_{\text{min}}$ (FG08: $P_{01}$) |

- **Effect**: Separates cold-montane **FG03 *Bombus*** (**33.41 °C**) from warm-tolerant **FG08** (**35.86 °C**) and **FG04 *Apis*** (**41.53 °C**); realized macroclimatic bounds ($\neq$ physiological $CT_{\text{max}}$; Araújo et al., 2013).

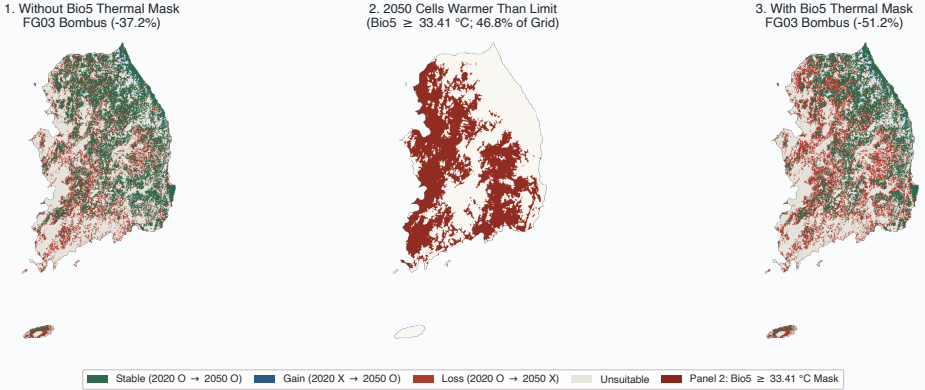
# [Step 2] Thermal Cutoffs

- **BAU-SSP585 `Bio5` (34.1 → 35.6 °C)**: Group cutoffs ($T_{max}$) exclude **46.8–53.8%** of cells in cool-adapted groups (**FG02, 03, 06, 07, 09, 12**) vs. **0.0–5.6%** in warm-tolerant groups (**FG01, 05, 10, 11**).



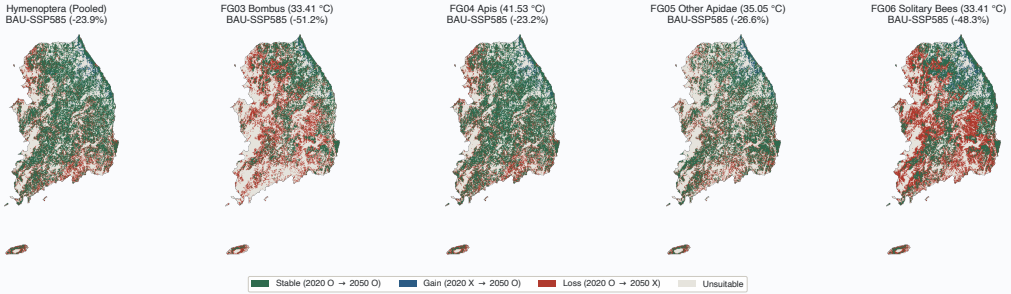
# [Step 2] Realized Upper Thermal Masks Expose Lowland Loss

- **Climatic Niche Cutoff (FG03 *Bombus*, `Bio5` $\geq$ 33.41 °C)**: Applying realized upper thermal masks sharpens BAU-SSP585 loss from -37.2% to **-51.2%**.



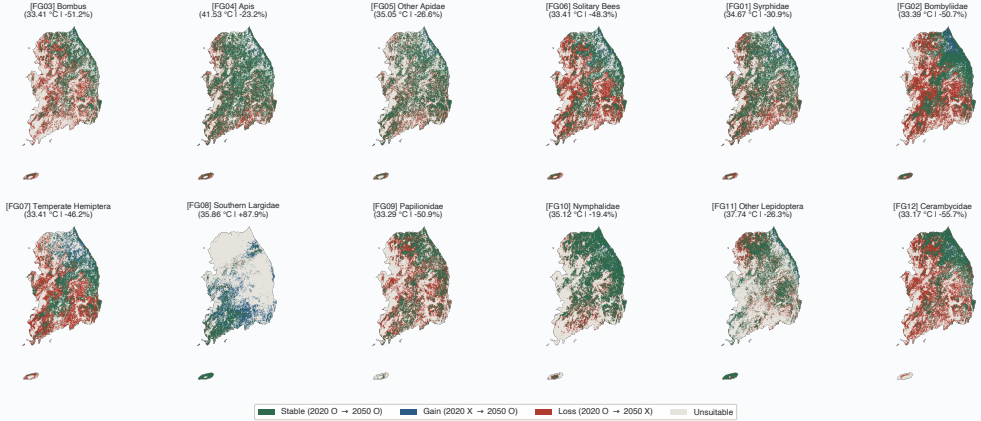
# [Step 2] Contrasting Responses Within Hymenoptera (FG03–FG06)

- Hymenoptera (-23.9%, BAU-SSP585) masks losses in **FG03 *Bombus* (-51.2%)** and **FG06 Solitary (-48.3%)** vs. **FG04 *Apis* (-23.2%)**.



# [Step 2] Range Shifts Across All 12 Functional Groups (FG01–FG12)

- Cool-adapted groups (**FG02, FG03, FG06, FG07, FG09, FG12**) lose **46–56%**, whereas subtropical **FG08** expands **+87.9%**.



# [Step 2] Quantitative Summary of 2050 Range Shifts (FG01–FG12)

- **Shift Metrics (BAU-SSP585)**: $\Delta\bar{z}$ = mean elevation shift (72–100% from lowland loss); $\Delta\bar{y}$ = northward centroid shift (**FG08**: $-1.2$ m, **+24.3 km**).

| Group | Order | Cells | AUC | $T_{\max}$ | 2020 Area | SSP126 | SSP585 | Elev. $\Delta\bar{z}$ | North $\Delta\bar{y}$ |
|---|---|---|---|---|---|---|---|---|---|
| **FG03** *Bombus* | Hym. | 927 | 0.628 | 33.4 °C | 43,201 km² | -14.9% | **-51.2%** | +80.9 m | +31.3 km |
| **FG04** *Apis* | Hym. | 2,352 | 0.625 | 41.5 °C | 54,100 km² | -6.3% | -23.2% | +39.5 m | +16.7 km |
| **FG05** Other Apidae | Hym. | 1,343 | 0.641 | 35.1 °C | 45,603 km² | -22.5% | -26.6% | +32.4 m | +16.1 km |
| **FG06** Solitary Bees | Hym. | 1,158 | 0.623 | 33.4 °C | 53,338 km² | -23.1% | **-48.3%** | +106.5 m | +33.8 km |
| **FG01** Syrphidae | Dip. | 2,753 | 0.620 | 34.7 °C | 50,667 km² | -6.2% | -30.9% | +57.5 m | +26.6 km |
| **FG02** Bombyliidae | Dip. | 519 | 0.616 | 33.4 °C | 57,951 km² | -18.9% | **-50.7%** | +139.1 m | +38.3 km |
| **FG07** Temp. Hemiptera | Hem. | 2,025 | 0.621 | 33.4 °C | 47,514 km² | -41.5% | **-46.2%** | +127.4 m | +49.9 km |
| **FG08** S. Largidae | Hem. | 124 | 0.777 | 35.9 °C | 12,660 km² | +128.3% | **+87.9%** | -1.2 m | **+24.3 km** |
| **FG09** Papilionidae | Lep. | 535 | 0.644 | 33.3 °C | 46,597 km² | -18.1% | **-50.9%** | +67.6 m | +17.8 km |
| **FG10** Nymphalidae | Lep. | 421 | 0.680 | 35.1 °C | 46,278 km² | +12.6% | -19.4% | +33.4 m | +17.9 km |
| **FG11** Other Lepidoptera | Lep. | 296 | 0.620 | 37.7 °C | 36,237 km² | +38.2% | -26.3% | +15.7 m | +1.1 km |
| **FG12** Cerambycidae | Col. | 782 | 0.642 | 33.2 °C | 53,089 km² | -20.3% | **-55.7%** | +136.5 m | +48.2 km |

# STEP 3. Plant–Pollinator Coupling & Spatial Mismatch

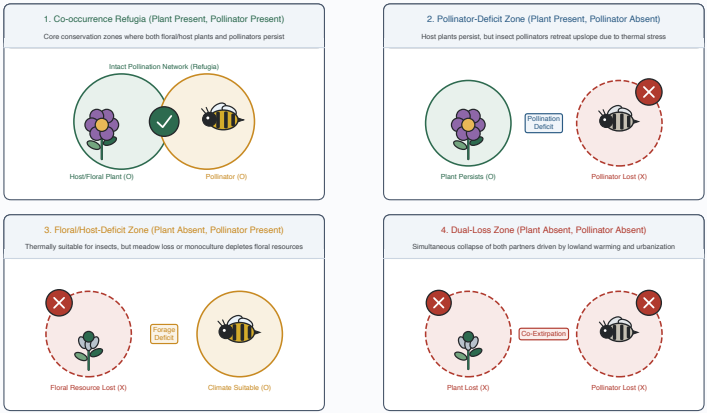
- **Biotic Decoupling Under Climate Change**:
  - Divergent range shifts erode spatial co-occurrence even in generalist networks (**Schweiger et al., 2008; Schleuning et al., 2016; Briggs & Anderson, 2026**); abiotic SDM risk alone poorly predicts secondary extinction risk ($r < 0.3$ under phenological mismatch; **Peng et al., 2025**)

| Regime | Pollinator | Plant | Ecological Consequence |
|---|---|---|---|
| **Co-occurrence refugia** | Suitable | Suitable | Shared climatic & floral persistence |
| **Pollinator deficit** | Unsuitable | Suitable | Plants persist; pollinators lost to heat |
| **Plant deficit** | Suitable | Unsuitable | Climate suitable; host/floral plants lost |
| **Dual loss** | Unsuitable | Unsuitable | Simultaneous lowland co-extirpation |

- **Objective**: Evaluate whether suitable insect climate space retains floral and host-plant overlap by 2050.

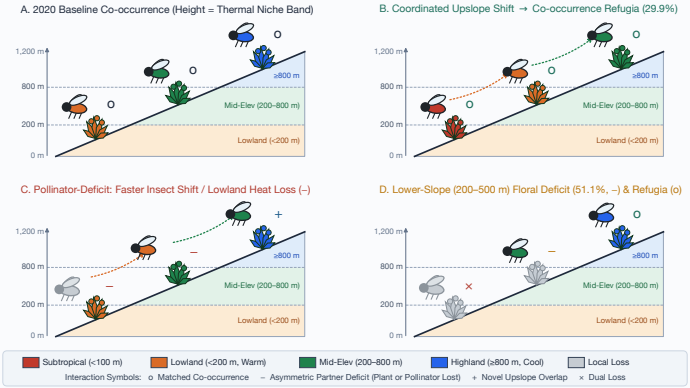
# [Step 3] Spatial Decoupling of Pollinators and Host/Floral Plants

- **Overlay**: Classifies 1 km cells into **Co-occurrence Refugia**, **Pollinator Deficit**, **Floral/Host Deficit**, and **Dual Loss** (cf. Schweiger et al., 2008; Schleuning et al., 2016; Briggs & Anderson, 2026).



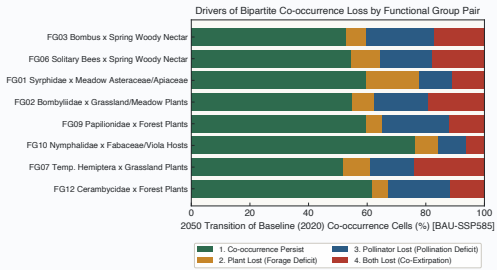
# [Step 3] Spatial Mismatch Regimes Along Elevational Gradients

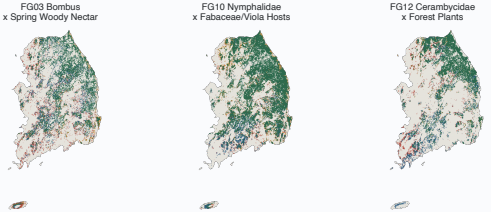
- Across lowland (<200 m), mid-slope (200–800 m), and highland ($\geq$ 800 m) bands, divergent range shifts (**Briggs & Anderson, 2026**) and host/floral loss (**Schweiger et al., 2008; Schleuning et al., 2016**) split baseline pairs (o) into deficits (–), novel overlap (+), and dual loss (×).



# [Step 3] Contraction of Shared Pollinator–Plant Suitable Habitats

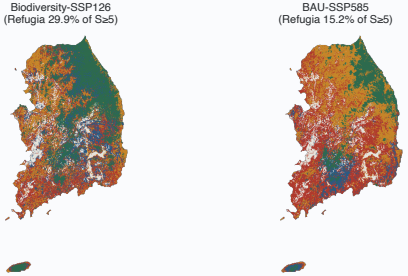
- **BAU-SSP585 Overlap (Schoener's $D$; Warren et al., 2008)**: Meadow loss (-23.9%) deepens **FG01 × Meadow** contraction (**-34.2%** vs. -30.9%); in **FG03 × Spring Woody** (**-43.0%**, $D$: 0.84 $\rightarrow$ 0.64), **23.1%** of 2020 shared cells lose *Bombus* (54.0% of 2050 plant cells lack *Bombus*).



# [Step 3] Lower-Slope Floral Deficits and Co-occurrence Refugia

- **Community Synthesis (274 Plants × 139 Pollinators; $S_{2020} \geq 5$):** **Biodiversity-SSP126** retains **29.9%** national refugia ($\Delta S \geq$ -20%) vs. **15.2%** in **BAU-SSP585** (**51.1%** floral/host deficit at **200–500 m**).



# STEP 4. Community Turnover & Ecological Synthesis

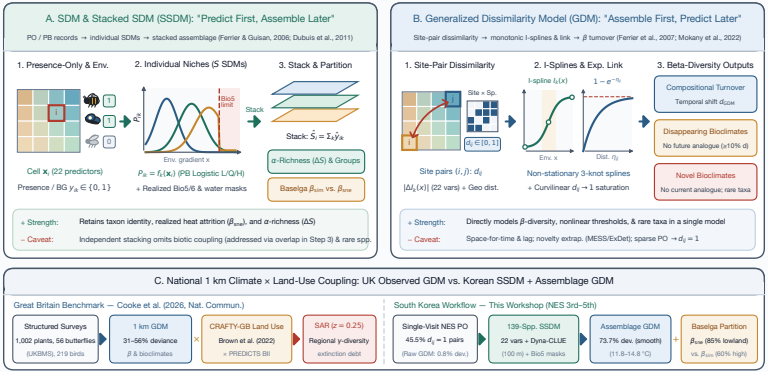
- **Framework Integration**
  - Integrates **SSDM** (Dubuis et al., 2011; Calabrese et al., 2014), **GDM** (Ferrier et al., 2007; Mokany et al., 2022), and **Baselga (2010)** partitioning ($\beta_{\text{sor}} = \beta_{\text{sim}} + \beta_{\text{sne}}$)

| Dimension | Stacked SDM (SSDM) | Generalized Dissimilarity Modelling (GDM) |
|---|---|---|
| **Main focus** | Species distributions & richness ($\Delta S$) | Nonlinear compositional turnover ($d_{ij}$) |
| **Response input** | Single-cell PO/PB ($y_{ik}$) $\rightarrow$ suitability $p_{ik} \in [0, 1]$ | Site-pair dissimilarity ($d_{ij} \in [0, 1]$, 25k pairs) |
| **Predictors** | 22 env. vars ($x_i$) + `Bio5/6` & water masks | 22 env. $\lvert\Delta I_k(x)\rvert$ + `Geo_Distance` spline |
| **Main output** | Richness & $\beta_{\text{sim}}$ / $\beta_{\text{sne}}$ maps | Monotonic I-spline turnover ($1 - e^{-\eta_{ij}}$) |
| **Ecological role** | Quantifies richness loss & nested attrition | Identifies critical turnover thresholds |

- **Objective**: Partition 2050 compositional change into lowland nestedness loss vs. highland species replacement.

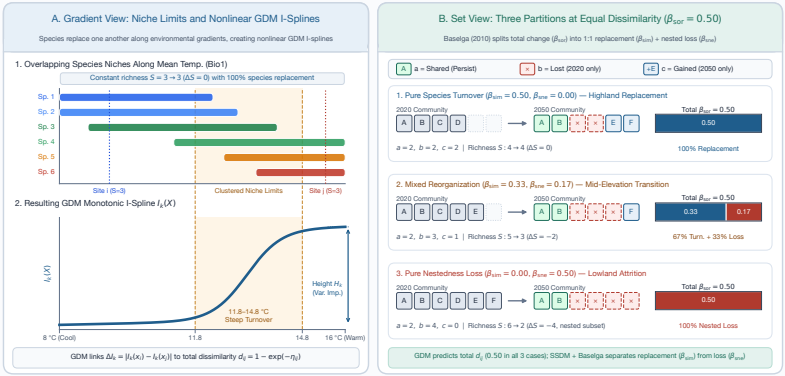
# [Step 4] Conceptual Comparison: SDM (Species Stacking) vs. GDM

- **Paradigms**: **SDM / SSDM** (*predict first, assemble later*; Ferrier & Guisan, 2006; Zurell et al., 2020) vs. **GDM** (*assemble first, predict later*; Ferrier et al., 2007).
- **National Coupling**: **Cooke et al. (2026, *Nat. Commun.*)** link UK 1 km GDMs with **CRAFTY-GB** (Brown et al., 2022); we couple **Dyna-CLUE (100 m)** with **SSDM + assemblage GDM**.



# [Step 4] Compositional Turnover: Gradient and Set-Partitioning Views

- **Gradient vs. Set Decomposition**: Clustered niche limits (**A**) produce steep nonlinear **GDM I-splines** even at constant richness ($\Delta S = 0$); equal total dissimilarity ($\beta_{sor} = 0.50$, **B**) spans **100% replacement** ($\beta_{sim}$) to **100% nested loss** ($\beta_{sne}$; Baselga, 2010).



# [Step 4] Meaning of Baselga (2010) Beta-Diversity Partitioning

- **Decomposition**: Total Sørensen dissimilarity ($\beta_{\text{sor}}$) conflates **1:1 species replacement** ($\Delta S = 0$) with **uncompensated species loss** ($\Delta S < 0$).
- **Cell Counts (2020 $\rightarrow$ 2050)**: Shared species $a$ (persist), lost species $b$ (2020 only), gained species $c$ (2050 only).

| Component | Formula | Ecological Meaning & Limiting Case |
|---|---|---|
| **Total ($\beta_{\text{sor}}$)** | $\frac{b+c}{2a+b+c} = \beta_{\text{sim}} + \beta_{\text{sne}}$ | Overall Sørensen dissimilarity (0: identical, 1: disjoint) |
| **Turnover ($\beta_{\text{sim}}$)** | $\frac{\min(b,c)}{a+\min(b,c)}$ | **Species replacement** (Simpson): 1:1 substitution (0 if $b = 0$ or $c = 0$) |
| **Nestedness ($\beta_{\text{sne}}$)** | $\beta_{\text{sor}} - \beta_{\text{sim}}$ | **Nested subset loss**: attrition without replacement ($= \beta_{\text{sor}}$ if $c = 0$) |

- **Role vs. GDM**: **GDM** models total $d_{ij}$ only, whereas **SSDM + Baselga** separates highland **replacement** from lowland **attrition** (cf. $\beta_{\text{rich}}$; Carvalho et al., 2012).

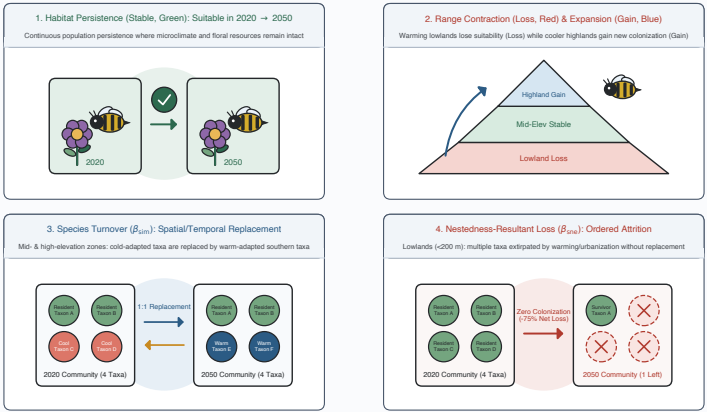
# [Step 4] Methodological Assumptions & Extrapolation Limits

- **Correlative Projection Caveats**: **SSDM** and **GDM** project *potential* suitability rather than realized communities, omitting abundance, competition, and phenology (Blois et al., 2013; Peng et al., 2025; Briggs & Anderson, 2026).

| Assumption / Issue | Limitation in SDM / GDM | Diagnostic / Mitigation |
|---|---|---|
| **Space-for-time** | Assumes equilibrium & unlimited dispersal | Potential $\neq$ realized (Fitzpatrick & Keller, 2015) |
| **Biotic & timing** | Omits trophic links & phenology | Overlap (Briggs & Anderson, 2026) |
| **Sparse PO** | Sparse PO $\rightarrow d_{ij} = 1$; SSDM smooths | Raw PO (0.8%) vs. plant (37%) & UK (Cooke et al., 2026) |
| **Novel climates** | Extrapolation past baseline bounds | `Bio5/6` masks + **MESS** / **ExDet** (Mesgaran et al., 2014) |

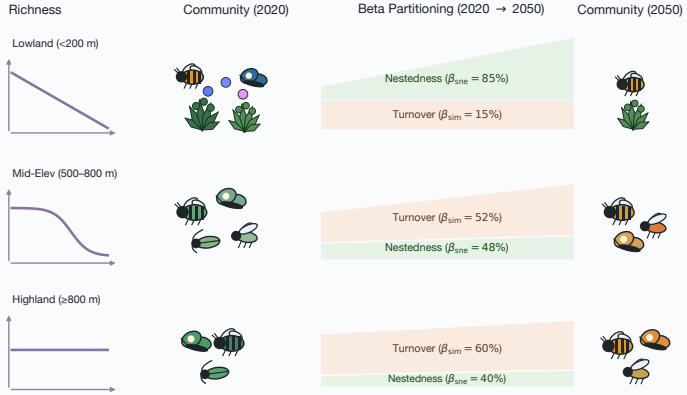
# [Step 4] Single-Taxon Range Shifts vs. Community Beta Partitioning

- **Species vs. Community (Baselga, 2010)**: Connects taxon **Stable / Loss / Gain** to highland **Species Turnover ($\beta_{sim}$)** and lowland **Nestedness-Resultant Loss ($\beta_{sne}$)**.



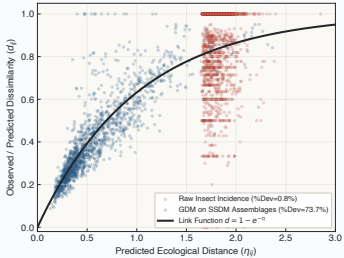
# [Step 4] Richness Decline, Nested Attrition, and Species Replacement

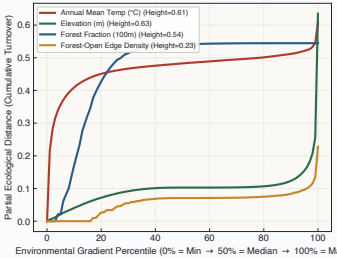
- **Elevational Gradient (SSDM + Baselga, 2010)**: Lowland (<200 m) **nestedness loss** ($\beta_{sne}$: **85.2%**) $\rightarrow$ mid-elevation (500–800 m) transition $\rightarrow$ highland ($\geq$ 800 m) **replacement** ($\beta_{sim}$: **59.7%**).

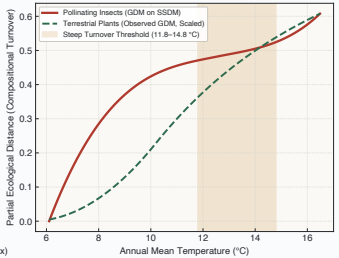


# [Step 4] Nonlinear Thermal Turnover Thresholds (GDM I-Splines)

- **Survey Completeness**: Sparse single-visit insect PO saturates ($d_{ij} = 1$: 45.5%; **0.81%** dev.) vs. observed plants (**37.04%**) and UK transects (31–56%; Cooke et al., 2026); **139-spp. SSDM-GDM** captures smoothed turnover (**73.70%**, steepest at **11.8–14.8 °C**).

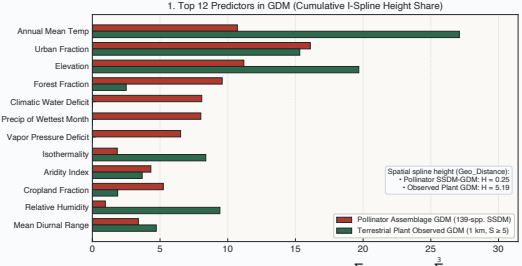


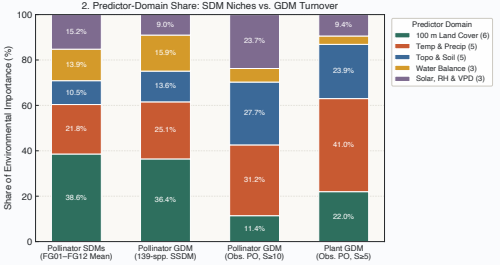



# [Step 4] SDM Weights vs. GDM Spline Heights

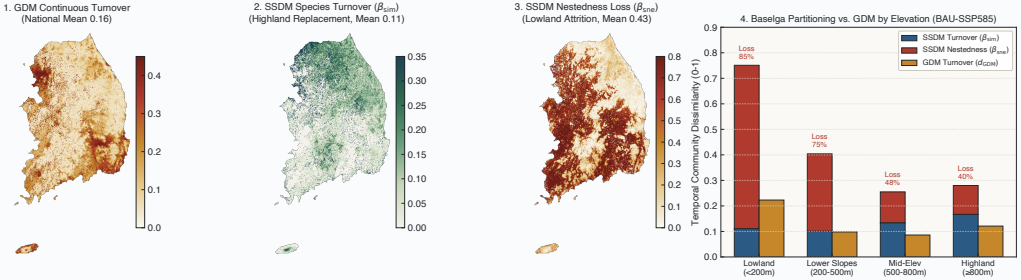
- **Importance Metrics**: **SDM** sums standardized $|\hat{\beta}|$ weights (land cover: **38.6%**), whereas **GDM** sums I-spline heights $H_k = \sum_{m=1}^{3} a_{km}$ (Ferrier et al., 2007; **36.4%** land cover in SSDM-GDM vs. **41.0%** climate in observed plants).



# [Step 4] Lowland Community Attrition vs. Highland Replacement

- **Baselga Partitioning (BAU-SSP585)**: Lowlands (<200 m) are dominated by **nestedness-resultant loss** ($\beta_{sne}$ = 0.640; **85.2%**), whereas highlands ($\geq$ 800 m) show **species replacement** ($\beta_{sim}$ = 0.167; **59.7%**).



# Summary

1. **Functional differentiation**
   - Group-level SDMs expose severe contractions masked by order pooling (**FG03 -51.2%**, **FG06 -48.3%**, **FG09 -50.9%**, **FG12 -55.7%**).
2. **Trophic & spatial connectivity**
   - Climate mitigation and habitat conservation (**Biodiversity-SSP126**) nearly double spatial co-occurrence refugia (**29.9%** vs. **15.2%** in **BAU-SSP585**), addressing interaction risks beyond single-taxon climate SDMs (Peng et al., 2025; Briggs & Anderson, 2026).
3. **Multidimensional community turnover**
   - **SSDM + GDM** (11.8–14.8 °C threshold) separates lowland nestedness attrition ($\beta_{sne}$: **85.2%**) from highland replacement ($\beta_{sim}$: **59.7%**).

- **Takeaway**: **Conserving pollinators requires more than climate space** — it requires protecting forest–open mosaics, host/floral plants, and spatial connectivity.

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**1. Presence–Background SDM, Spatial Block CV, & Realized Thermal Limits**: Phillips & Dudík (2008) *Ecography* 31:161–175 · Phillips et al. (2009) *Ecol. Appl.* 19:181–197 · Elith et al. (2011) *Divers. Distrib.* 17:43–57 · Renner & Warton (2013) *Biometrics* 69:274–281 · Allouche et al. (2006) *J. Appl. Ecol.* 43:1223–1232 (TSS) · Roberts et al. (2017) *Ecography* 40:913–929 (spatial block CV) · Araújo et al. (2013) *Ecol. Lett.* 16:1206–1219 · Fick & Hijmans (2017) *Int. J. Climatol.* 37:4302–4315 (`WorldClim 2`) · Colwell et al. (2008) *Science* 322:258–261 · Elsen & Tingley (2015) *Nat. Clim. Change* 5:772–776.

**2. Plant–Pollinator Spatial/Phenological Mismatch & Secondary Extinction Risk**: **Briggs, S. D., & Anderson, J. T. (2026)**. Spatial mismatches of plant–pollinator interactions under future climate conditions. *J. Biogeogr.*, 53(4), e70204 · **Peng, S., Ellison, A. M., & Davis, C. C. (2025)**. Climate change intensifies plant–pollinator mismatch and increases secondary extinction risk for plants in northern latitudes. *PNAS*, 122(40), e2506265122 · Schweiger et al. (2008) *Ecology* 89:3472–3479 · Schleuning et al. (2016) *Nat. Commun.* 7:13965 · Warren et al. (2008) *Evolution* 62:2868–2883 (Schoener's $D$).

**3. Stacked SDM (SSDM), GDM, & Beta-Diversity Partitioning**: Ferrier & Guisan (2006) *J. Appl. Ecol.* 43:393–404 · Dubuis et al. (2011) *Divers. Distrib.* 17:1122–1131 · Calabrese et al. (2014) *Glob. Ecol. Biogeogr.* 23:99–112 · Zurell et al. (2020) *J. Biogeogr.* 47:101–113 · Ferrier et al. (2007) *Divers. Distrib.* 13:252–264 (GDM) · Mokany et al. (2022) *Glob. Ecol. Biogeogr.* 31:802–821 · Baselga (2010) *Glob. Ecol. Biogeogr.* 19:134–143 ($\beta_{\text{sor}} = \beta_{\text{sim}} + \beta_{\text{sne}}$) · Carvalho et al. (2012) *Glob. Ecol. Biogeogr.* 21:760–771.

**4. Scenario Coupling, Space-for-Time Substitution, & Extrapolation Novelty**: **Cooke, R. et al. (2026)**. Future scenarios for British biodiversity under climate and land-use change. *Nat. Commun.*, 17, 2704 · Brown et al. (2022) *Earth's Future* 10:e2022EF002905 (`CRAFTY-GB`) · Blois et al. (2013) *PNAS* 110:9374–9379 · Fitzpatrick & Keller (2015) *Ecol. Lett.* 18:1–16 · Williams & Jackson (2007) *Front. Ecol. Environ.* 5:475–482 · Elith et al. (2010) *Methods Ecol. Evol.* 1:330–342 (`MESS`) · Mesgaran et al. (2014) *Divers. Distrib.* 20:1147–1159 (`ExDet`).